

Engineering Mathematics: Matrix Reduction to Normal Form

1. Problem Statement

Find the normal form and the rank of the matrix:

$$A = \begin{bmatrix} 8 & 1 & 3 & 6 \\ 0 & 3 & 2 & 2 \\ -8 & -1 & -3 & 4 \end{bmatrix}$$

2. Transformation Process

Step 1: Establish a Leading Pivot

To make the arithmetic easier, we swap C_1 and C_2 to bring the element **1** to the $(1, 1)$ position.

$$\begin{bmatrix} 8 & 1 & 3 & 6 \\ 0 & 3 & 2 & 2 \\ -8 & -1 & -3 & 4 \end{bmatrix} (C_1 \leftrightarrow C_2) \longrightarrow \begin{bmatrix} 1 & 8 & 3 & 6 \\ 3 & 0 & 2 & 2 \\ -1 & -8 & -3 & 4 \end{bmatrix}$$

Step 2: Clear the First Column

We use the pivot at a_{11} to eliminate the elements below it.

Apply: $R_2 \rightarrow R_2 - 3R_1$ and $R_3 \rightarrow R_3 + R_1$.

$$\begin{bmatrix} 1 & 8 & 3 & 6 \\ 3 & 0 & 2 & 2 \\ -1 & -8 & -3 & 4 \end{bmatrix} (R_2 - 3R_1, R_3 + R_1) \longrightarrow \begin{bmatrix} 1 & 8 & 3 & 6 \\ 0 & -24 & -7 & -16 \\ 0 & 0 & 0 & 10 \end{bmatrix}$$

Step 3: Clear the First Row

Now we use column operations to clear the elements to the right of the pivot.

Apply: $C_2 \rightarrow C_2 - 8C_1$, $C_3 \rightarrow C_3 - 3C_1$, $C_4 \rightarrow C_4 - 6C_1$.

$$\begin{bmatrix} 1 & 8 & 3 & 6 \\ 0 & -24 & -7 & -16 \\ 0 & 0 & 0 & 10 \end{bmatrix} (C_2, C_3, C_4 \text{ ops}) \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -24 & -7 & -16 \\ 0 & 0 & 0 & 10 \end{bmatrix}$$

Step 4: Normalize the Second and Third Pivots

We need to isolate the remaining non-zero elements. We swap C_2 and C_4 to bring the value **10** into a more manageable position, then normalize the rows.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -24 & -7 & -16 \\ 0 & 0 & 0 & 10 \end{bmatrix} (C_2 \leftrightarrow C_4) \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -16 & -7 & -24 \\ 0 & 10 & 0 & 0 \end{bmatrix}$$

By continuing elementary operations ($R_2 \rightarrow R_2 + 2R_3$, then clearing the remaining constants), the matrix reduces to its canonical identity structure.

Step 5: Initial State and Pivot 1

The first pivot a_{11} is already 1, and its column is cleared.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -16 & -7 & -24 \\ 0 & 10 & 0 & 0 \end{bmatrix} (R_2 \leftrightarrow R_3) \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 10 & 0 & 0 \\ 0 & -16 & -7 & -24 \end{bmatrix}$$

Step 6: Establishing Pivot 2

We normalize the second row to get a leading 1 at a_{22} .

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 10 & 0 & 0 \\ 0 & -16 & -7 & -24 \end{bmatrix} \left(R_2 \rightarrow \frac{1}{10} R_2 \right) \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -16 & -7 & -24 \end{bmatrix}$$

Step 7: Vertical Clearing (Row Operations)

Use the new pivot at a_{22} to clear the value below it in R_3 .

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -16 & -7 & -24 \end{bmatrix} (R_3 \rightarrow R_3 + 16R_2) \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -7 & -24 \end{bmatrix}$$

Step 8: Establishing Pivot 3

We normalize the third row to get a leading 1 at a_{33} .

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -7 & -24 \end{bmatrix} \left(R_3 \rightarrow -\frac{1}{7} R_3 \right) \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{24}{7} \end{bmatrix}$$

Step 9: Horizontal Clearing (Column Operations)

Finally, we use column operations to eliminate any remaining values to the right of our pivots to reach the identity structure.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{24}{7} \end{bmatrix} \left(C_4 \rightarrow C_4 - \frac{24}{7} C_3 \right) \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

10. Final Result

The matrix is now in the form $[I_3 \mid 0]$.

- **Rank of the Matrix (r):** 3
- **Canonical Form:**

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Which is represented as:

$$\begin{bmatrix} I_3 & 0 \end{bmatrix}$$

4. Conclusion

- **Normal Form:** $\begin{bmatrix} I_3 & 0 \end{bmatrix}$
- **Rank:** The number of non-zero rows is 3.
- **Result:** $\rho(A) = 3$